

Introduction

The aim of this analysis is to classify human activities such as sitting, standing, laying, walking, walking upstairs and walking downstairs based on smartphone sensor measurements. The data originate from accelerometer and gyroscope signals collected with a Samsung smartphone during different activities. Recognizing activities from mobile sensor data is relevant because smartphones are widely available and can be used for health monitoring, fitness tracking or context-aware applications. Therefore, the main goal is to build a reliable prediction model and to estimate how well it performs on unseen persons.

Methods

The dataset was created by Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge L. Reyes-Ortiz as part of a research project on Human Activity Recognition (HAR) using smartphone sensors. It was published in 2013 in the paper “A Public Domain Dataset for Human Activity Recognition Using Smartphones” at ESANN 2013. The data was collected from 30 volunteers aged between 19 and 48 years. They performed six daily activities while carrying a Samsung Galaxy S II smartphone on their waist, whose accelerometer and gyroscope recorded motion data at a sampling rate of 50 Hz. The dataset is publicly available through the UCI Machine Learning Repository. ^[1]

The original dataset contained both raw sensor signals and preprocessed feature vectors. The data was divided into 70% training and 30% test sets. In each observation, there were 561 features, 1 grouping feature (corresponding to the subject from whom this data was recorded), and 1 “activity” label. The 561 features include time-domain and frequency-domain statistics such as means, standard deviations, signal magnitudes, entropy measures, and frequency characteristics.

The data samples we were provided were adapted to our task. The raw sensor signals have been removed, and only the following files remain:

- `measures.csv` : Training set containing 7352 labeled observations
- `to_predict.csv` : Blind-test set containing 2947 unlabeled observations
- `features_info.txt` : Explanation file about the features
- `README.txt` : General explanation file about the data and its source

In total, 10299 observations were provided.

The label corresponds to the activity the subjects were performing at the time the data was captured. In total, there were 6 distinct activities, and therefore 6 classes:

1. WALKING
2. WALKING_UPSTAIRS
3. WALKING_DOWNSTAIRS
4. SITTING
5. STANDING
6. LAYING

Each row represents one observation window of sensor data. The continuous accelerometer and gyroscope signals were segmented into fixed-width windows of 2.56 seconds (128 readings at 50 Hz) with 50% overlap between consecutive windows. For each window, 561 features were extracted and stored as a single feature vector. Therefore, one line in the CSV file corresponds to a single 2.56-second activity window and its associated activity label.

Data preparation and exploratory analysis

There are no duplicate columns or rows, and no missing values in either of the data sets provided. All features lie strictly between -1 and 1.

There are some features that don't span the entire [-1, 1]-range. The most extreme example is `tGravityAcc-max()-Y`, which only spans the range [-0.49, 0.97]. However, this does not indicate any sort of error with the data – most features represent minimums and maximums of specific values, and these naturally tend toward one of the extremes, making their range of values expectedly smaller. In any case, 90% of the features span at least 95% of the range [-1, 1].

There do not seem to be any specific anomalies. No entries with abnormally high or low values could be found. We used a Z-score outlier test, but even high values for Z gave back an enormous amount of pseudo-outliers, which pointed towards the methods ineffectiveness on this data. We have come to the conclusion that the outliers are insignificant or not present.

Splitting

To estimate the final error of our model, we split the entire data set into a 90% training set and 10% testing set. We used `StratifiedGroupKFold` to achieve that while preserving group coherence and class label distributions (it seems as if that's the only splitter that supports both these features). Technically, this function creates 10 distinct splits and is intended for use in cross-validation, but we just took the first split it created – thereby partitioning 1/10 of the data for the test set and 9/10 for the train set.

Feature analysis

To analyze feature correlation with the class, we used mutual information scores. We found that there are several features that correlate strongly with the activity. The top 10 are:

<code>tBodyAcc-max()-X</code>		1.006893
<code>tGravityAcc-min()-Y</code>		0.944219
<code>tBodyAccJerk-max()-X</code>		0.937857
<code>tGravityAcc-max()-Y</code>		0.928346
<code>tBodyAccMag-max()</code>		0.904677
<code>tGravityAccMag-max()</code>		0.904643
<code>tBodyAccJerk-max()-Y</code>		0.892837
<code>fBodyAcc-std()-X</code>		0.890931
<code>fBodyAcc-bandsEnergy()-1,16</code>		0.890793
<code>fBodyAcc-bandsEnergy()-1,8</code>		0.887700

Each of these features measure something about acceleration. It would seem that measurements from the gyroscope are far less important for prediction than those of the accelerometer.

To learn more about these correlations, we took the 3 features which correlated the most with the activity, and analyzed their box plots with respect to different activity classes.

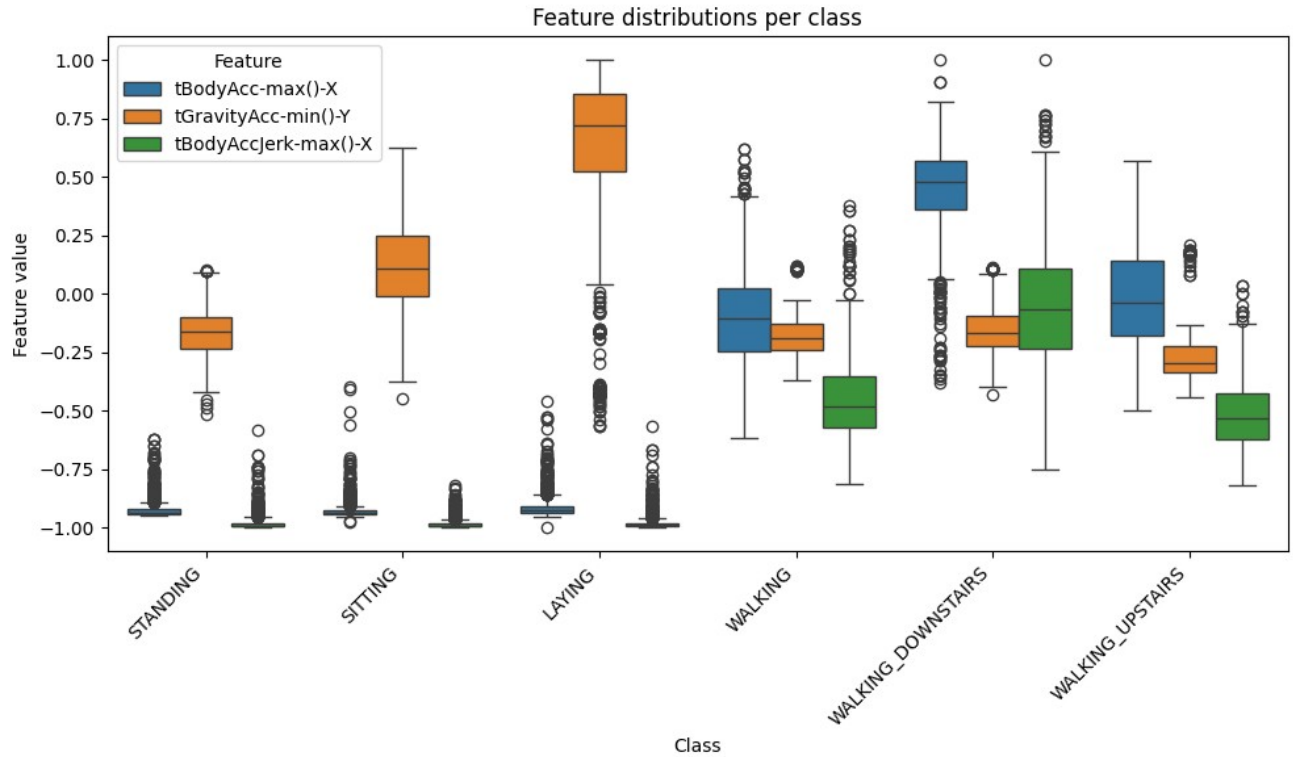


Fig. 1: Box plots of the three features most strongly correlated with the activity label, separated by each activity class.

As can be seen on figure 1, the distributions of each of these features for each of the classes are unique. The correlation is, therefore, definitely not accidental. These features are significant for the final classification.

Two groups of activities can be spotted on this graph: **static** and **dynamic** activities.

Static activities (STANDING, SITTING and LAYING) have a few common characteristics: namely, tBodyAcc-max()-X and tBodyAccJerk-max()-X take on values near -1 and have a small variance, while tGravityAcc-min()-Y has a larger variance and takes on values somewhere between -0.25 and 1.

Dynamic activities (WALKING, WALKING_DOWNSTAIRS and WALKING_UPSTAIRS) also have common characteristics: tBodyAcc-max()-X and tBodyAccJerk-max()-X have a high variance and take on values between -0.75 and 0.75, while tBodyAcc-max()-Y takes on values around -0.25 and has a small variance.

These distinctive characteristics in the groups are expected, as static activities don't involve motion, while dynamic activities do.

Correlations between features

To analyze how strong the correlations between features can be, we picked 9 most correlated feature pairs and created scatter plots for these pairs. As seen on figure 2, these are rather thin straight lines, implying that these features correlate extremely strongly and predictably.

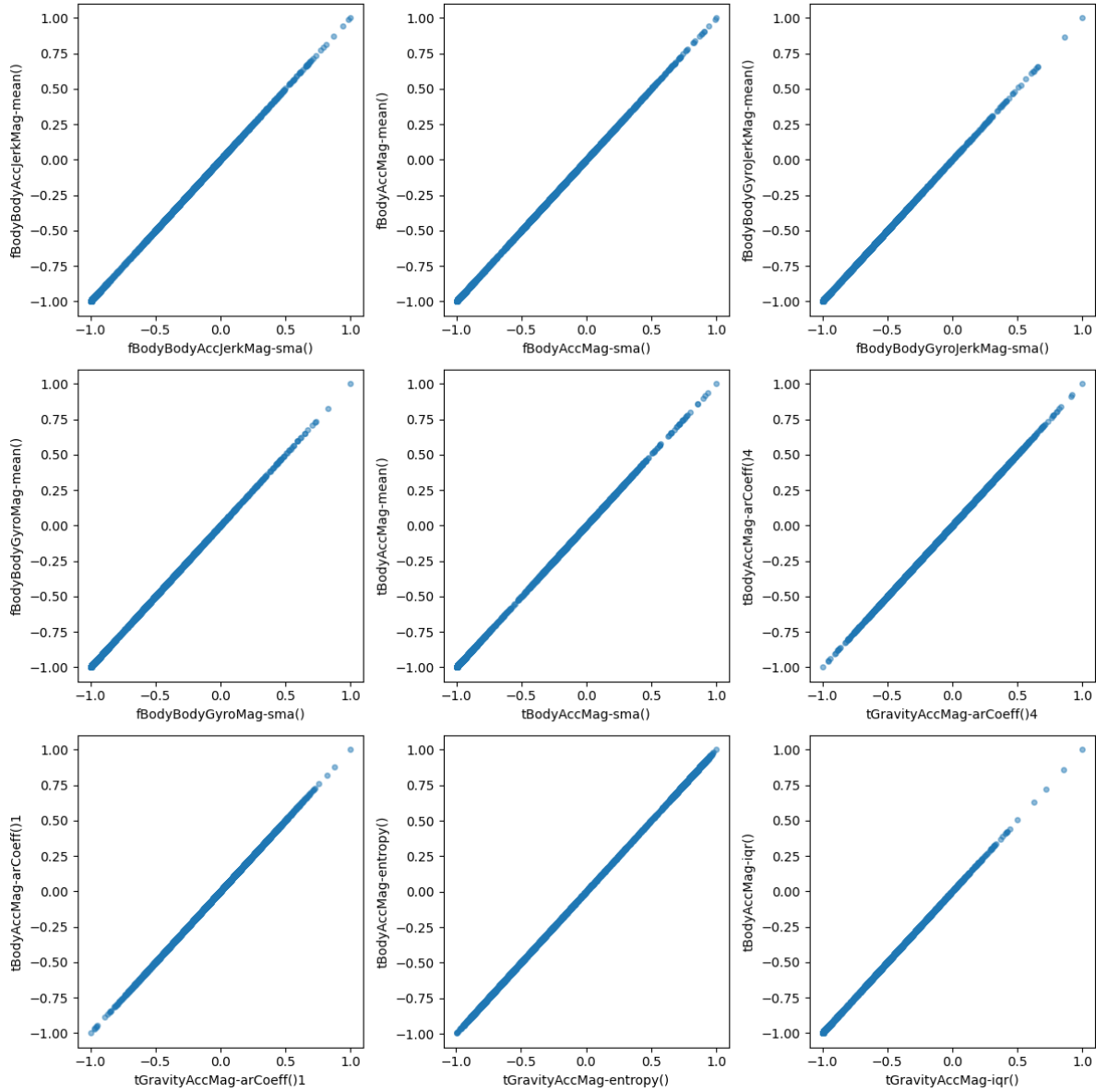


Fig. 2: Scatter plots of the 9 most inter-correlated feature pairs

We came to the conclusion that reducing the feature space is necessary before proceeding. We decided to reduce feature pairs whose correlation coefficient was at least 0.95. For each feature pair, we kept the feature which appeared earlier in the columns, and removed the other one automatically. In total, 308 features were removed, which is 55% of the features.

Model selection and error estimation

Model selection

We tested three different model setups with 5-fold stratified, group-aware cross-validation. For each model setup, we recorded the accuracy for each CV-fold and recorded the minimum, mean, and maximum of those scores, as well as the calculation time. The results are shown in the table below.

Classifier	Parameters	Min. accuracy	Mean accuracy	Max. accuracy	Calculation time
DecisionTreeClassifier		78.5%	83.6%	90.5%	3.4 seconds
RandomForestClassifier		85.1%	91.2%	97.9%	16.2 seconds
LogisticRegression	max_iter=1000	85.6%	92.3%	98.2%	46.3 seconds

LogisticRegression gave the best results out of the models we have tested, which is why we decided to use it for our model.

Model evaluation

To estimate model accuracy on previously unseen data, we applied the 90% model on the test set and used `accuracy_score` and `confusion_matrix` from the `sklearn.metrics` package to generate the accuracy and confusion matrix of the prediction. The results are as follows:

Accuracy on the test set: 95.6%

Confusion matrix	LAYING	SITTING	STANDING	WALKING	WALKING_DO WNSTAIRS	WALKING_UP STAIRS
LAYING	160	0	0	0	0	0
SITTING	0	114	0	0	0	0
STANDING	0	33	148	0	0	0
WALKING	0	0	0	117	1	0
WALKING_DO WNSTAIRS	0	0	0	0	105	0
WALKING_UP STAIRS	0	0	0	0	1	112

While the accuracy is within expected margins and doesn't deviate strongly from the CV error, the confusion matrix gives us an important piece of information. As can be seen, there are hardly any errors, except for a systematic misclassification error, which is highlighted, and which accounts for 94% of all misclassification errors. The class SITTING is consistently confused with STANDING (although not the other way around!). The activity SITTING is misclassified as STANDING in 22% of cases, and in our example 18% of STANDING classifications should actually be SITTING. This is certainly an error of the model we chose, because such a consistent misclassification error was not observed in other research papers.^[2]

Prediction

We created and used the 100% model in the following steps:

1. Load the training data
2. Drop the grouping feature "subject" from the train set (this feature is only relevant for avoiding data contamination, especially during CV; including it in the actual training data would bias the model in any case)
3. Find feature pairs that have an absolute correlation of higher than 95%
4. Reduce those feature pairs by dropping features that appear in later columns for each pair; record the columns dropped
5. Train a LogisticRegression model with `max_iter=1000` on the train set
6. Load the testing data
7. Drop the grouping feature "subject" and the features we dropped in step 4 from the test set
8. Predict the labels using `model.predict()` on the test set

Results

The 10 most important features all measured acceleration, pointing towards the major importance of acceleration for prediction. This tendency is supported by the coefficients of the 100% model, which strongly favor acceleration-based features.

Correlations between features were the most major problem with the data and constituted the majority of our data preparation. Feature reduction was necessary, and as a result 55% of all features were dropped.

Out of the three models we tested – DecisionTreeClassifier, RandomForestClassifier, and LogisticRegression – LogisticRegression gave the best accuracy results. That was the reason we chose to use it as our model in the end. Our model performs very well, with an estimated 95.6% accuracy. However, as we found out using the confusion matrix, it consistently misclassifies around 22% of SITTING cases as STANDING. Therefore, whenever the model returns STANDING as the activity, it is necessary to remain skeptical – there is a good chance the true activity is SITTING. On the other hand, its other classifications can be trusted to a very high degree.

Conclusions

This study investigated the classification of human activities using smartphone sensor measurements. Feature reduction was performed to avoid high-correlating feature pairs. Several machine learning models were evaluated using cross-validation, and the best-performing model, LogisticRegression, was selected for final testing. The results demonstrate that activity recognition can be performed with high accuracy using the provided sensor features.

The final model achieved strong performance on previously unseen subjects, indicating good generalization ability. However, a consistent class confusion issue was observed, indicating a potential limitation of the approach. A major shortcoming in this process was finding outliers – we found that Z-score outlier tests were completely ineffective at catching real outliers in this dataset. Still, the results are limited to the available dataset and may not fully represent real-world conditions. Future work could explore additional feature engineering techniques, outlier-finding tests, alternative machine learning algorithms, or deep learning approaches to further improve classification performance and robustness, as well as rectifying the misclassification problem.

References

- [1] Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L. (2013). *A Public Domain Dataset for Human Activity Recognition Using Smartphones*.
<https://www.esann.org/sites/default/files/proceedings/legacy/es2013-84.pdf>
- [2] Erhan Bulbul, Aydın Çetin, I. Dogru (2018). *Human Activity Recognition Using Smartphones*.
<https://www.semanticscholar.org/paper/Human-Activity-Recognition-Using-Smartphones-Bulbul-%C3%87etin/4c70cc7f44bfd0105168acb8d85f0c71718a579f>

See the “Figures” section. Confusion matrices of other classifiers don’t exhibit the same behavior.